Buderus Cold Work Tool Steel 2550

	C	Si	Mn	P	S	Cr	V	W
Typical analysis	0.60	0.70	0.30	0.025	0.010	1.10	0.15	2.00
Chemical composition as per SEL	0.55–0.65	0.70–1.00	0.15–0.45	≤ 0.030	≤ 0.030	0.90–1.20	0.10–0.20	1.70–2.20

Figures in % by mass

Register of European Steels (SEL)	60 WCrV 8
DIN EN ISO 4957	60 WCrV 8
AFNOR	55 WC 20
BS	~ BS 1

Characteristics
Good toughness, high fatigue strength, good cutting edge retention.

Applications
Blanking dies for thicker material, shearing blades, highly stressed punch dies, woodworking tools, cutters and choppers, ejectors.

Delivered condition
Annealed to max. 229 HB

Physical properties (reference values)

Thermal expansion coefficient (10^{-6}/K)	20–100 °C 11.0	20–250 °C 12.3	20–500 °C 13.8
Thermal conductivity (W/mK)	20 °C 37.0	250 °C 38.0	500 °C 36.0
Young's modulus (GPa)	20 °C 210	250 °C 196	500 °C 176

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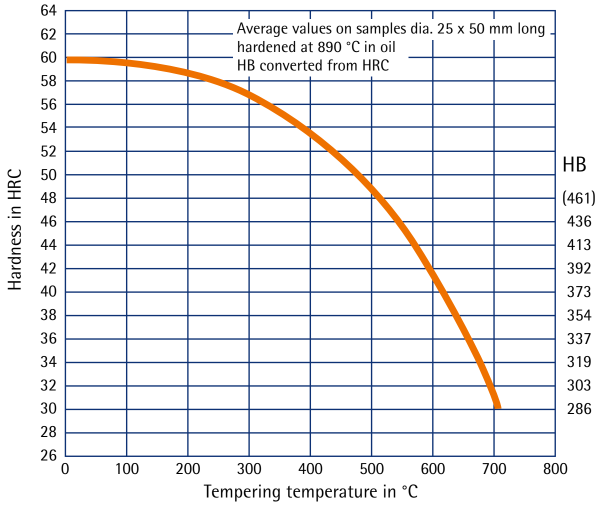
Heat treatment		
Stress relieving	Temperature: Duration: Cooling:	Approx. 650 °C in the annealed state 1 hour per 50 mm wall thickness Furnace
Soft annealing	Temperature: Duration: Cooling:	760 °C 1 hour per 25 mm wall thickness Furnace
Hardening	Temperature: Duration:	890 °C 1 minute per mm wall thickness
Quenching hardness	Max. 60 HRC	in oil, hot bath or vacuum
Tempering	Temperature: Duration: Cooling:	See tempering curve 1 hour per 25 mm wall thickness Air
Working hardness	56–60 HRC	

Tempering curve



TTT curve (continuous)